Lecturer: Flemming Christensen

INSTITUT FOR ELEKTRO- OG COMPUTERTEKNOLOGI, AARHUS UNIVERSITET

Email: flech@ece.au.dk

Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed., https://www.probabilitycourse.com

Course: Stokastisk modellering og behandling (SWSTS-01)

Conditional Probability

Topic: Updating probabilities with new information

- In real life, we often revise what we believe when we get new data.
- **Key question**:
  How does new information change the probability of an event?

Example: Weather

Imagine a city where:

- 23% of the days are rainy.
- If we pick a random day:

$$P(R) = 0.23$$

where R = "it rains on the chosen day"

New Information Appears:

- This time, we also know **it's cloudy**.

*What's the probability that it rains **given** that it's cloudy?*

Conditional Probability:

$$P(R \mid C)$$

- This is read as: "**Probability of rain given that it is cloudy**"
- This is called the **conditional probability**

Conditional Probability

Formal Definition and Formula

- For events A and B with $P(B) > 0$, the conditional probability of A given B is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- **Interpretation**: When B is known to happen, we "restrict" the sample space to B, and ask:

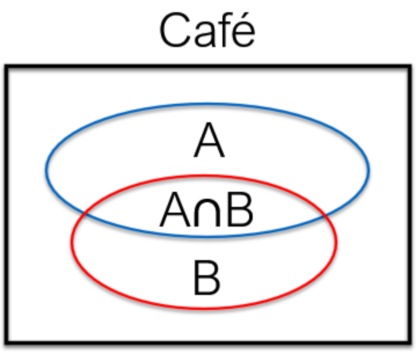
"How likely is A inside B?"

Example: Café

What is the probability that a customer orders coffee given they ordered cake?

Events:

- **A** = Customer orders **coffee**
- **B** = Customer orders **cake**



$$\Pr(A) = 0{,}70$$

$$\Pr(B) = 0{,}40$$

$$\Pr(A \cap B) = 0{,}20$$

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)} = \frac{0{,}20}{0{,}40} = \frac{1}{2}$$

Conditional Probability

Example: Debugging a Simple Program

A program logs the result of two checks:

- C_1: result of **Check 1**
- C_2: result of **Check 2**

Each check returns a value from the set {0, 1, 2} (like error codes or flags).

You observe that the **sum** of the checks is **2**.

What's the probability that **at least one check returned 1**?

Events given:

- Let A: "**Check 1 = 1 or Check 2 = 1**"
- Let B: "**Check 1 + Check 2 = 2**"

Sample Space:

$$S = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2)\}$$

Outcomes for Each Event:

- **Event A (at least one check is 1)**:

$$A = \{(1,0), (1,1), (1,2), (0,1), (2,1)\}$$

- **Event B (sum = 2)**:

$$B = \{(0,2), (1,1), (2,0)\}$$

- **Intersection $A \cap B$**:

$$A \cap B = \{(1,1)\}$$

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{1}{3}$$

Quiz

In a lab testing microcontrollers (MCUs), the following data is collected:

- 42% of all MCUs are **AVR** and **Stable**.
- 74% of all MCUs are **Stable**.

If an MCU is found to be Stable, what is the probability that it is an AVR?

A. 42%

B. 50%

C. 57%

D. 70%

Quiz

If event B is a subset of event A (i.e. $B \subset A$), what is the conditional probability $P(A \mid B)$?

A. $\frac{P(A)}{P(B)}$

B. $\frac{P(B)}{P(A)}$

C. 0

D. 1

Conditional Probability

Conditional Probability and Probability Axioms

Conditional probability is itself a **probability measure**, which means it satisfies the same axioms as ordinary probability. Specifically:

Axiom 1:

For any event A,

$$P(A \mid B) \geq 0$$

Axiom 2:

The conditional probability of event B given itself is 1:

$$P(B \mid B) = 1$$

Axiom 3 (Additivity for Disjoint Events):

If $A_1, A_2, A_3, \ldots$ are mutually disjoint events, then:

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots \mid B) = P(A_1 \mid B) + P(A_2 \mid B) + P(A_3 \mid B) + \cdots$$

These rules confirm that conditional probability behaves just like regular probability within the new "conditional" sample space (i.e., given that B has occurred).

Equivalent formulas

	Set Theory	Ordinary Probability	Conditional Probability
Complement Rule	$\lvert A^c \rvert = \lvert S \rvert - \lvert A \rvert$	$P(A^c) = 1 - P(A)$	$P(A^c \mid C) = 1 - P(A \mid C)$
Union Rule *(Inclusion - Exclusion)*	$\lvert A \cup B \rvert = \lvert A \rvert + \lvert B \rvert - \lvert A \cap B \rvert$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	$P(A \cup B \mid C) = P(A \mid C) + P(B \mid C) - P(A \cap B \mid C)$

Conditional Probability

chain rule for conditional probability:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_2, A_1) \cdots P(A_n \mid A_{n-1}, \ldots, A_1)$$

Example: Picking Non-Defective Products

A factory have **100 units**, and **5 are defective**. We pick **3 units at random**.

What is the probability that **all 3 are good**?

Solution

Let A_1, A_2, and A_3 be the events that the **1st**, **2nd**, and **3rd** units are **good**.

Use the Chain Rule

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_2, A_1)$$

- There are 95 good out of 100 units:

$$P(A_1) = \frac{95}{100}$$

- If first is good, 94 good remain out of 99:

$$P(A_2|A_1) = \frac{94}{99}$$

- If first two are good, 93 good remain out of 98:

$$P(A_3|A_2, A_1) = \frac{93}{98}$$

Result:

$$P = \frac{95}{100} \cdot \frac{94}{99} \cdot \frac{93}{98} = 0.8560$$

Conditional Probability

Example

A standard deck has **52 cards**:

- **26 red cards**
- **26 black cards**

You draw **4 cards at random without replacement.**

What is the probability that:

- The **first 3 cards are red**
- The **4th card is black?**

Solution

We'll compute:

$$P(R_1 \cap R_2 \cap R_3 \cap B_4) = P(R_1) \cdot P(R_2|R_1) \cdot P(R_3|R_1, R_2) \cdot P(B_4|R_1, R_2, R_3)$$

- $P(R_1) = \frac{26}{52}$
- $P(R_2|R_1) = \frac{25}{51}$
- $P(R_3|R_1, R_2) = \frac{24}{50}$
- $P(B_4|R_1, R_2, R_3) = \frac{26}{49}$

Result

$$P = \frac{26}{52} \cdot \frac{25}{51} \cdot \frac{24}{50} \cdot \frac{26}{49} \approx 0.0624$$

Quiz

From a standard **52-card deck**:

- 26 red cards
- 26 black cards

You draw **4 cards without replacement.**

What is the probability that:

- The **1st card is black**
- The **2nd, 3rd, and 4th cards are red**?

A) 0.8560

B) 0.0624

C) 0.1176

D) 0.2496

Independence

Definition

Two events A and B are **independent** if:

$$P(A \cap B) = P(A) \cdot P(B)$$

In case of independence, conditional probability simplifies:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

This holds when A and B are independent and $P(B) > 0$.

Disjoint ⇒ Not Independent

If A and B are **disjoint** and $P(A) > 0$, $P(B) > 0$, then:

$$P(A \cap B) = 0 \neq P(A)P(B)$$

Independence

NB. Independence / dependence is defined only for events that belong to the **same sample space**.

Example : single die roll

Let

$$\Omega = \{1,2,3,4,5,6\}, \quad A = \{1\}, \quad B = \{2\}.$$

A and B are events in the **same sample space, but they are disjoint**

So:

$$P(A \cap B) = 0 \neq \tfrac{1}{6} \cdot \tfrac{1}{6}$$

❌ **They are not independent.**

So in a **single roll**, saying "$A = \{1\}$ and $B = \{2\}$ are independent" is **false**.

Example: Two die rolls (same *joint* sample space)

Let

$$\Omega = \{1,\dots,6\} \times \{1,\dots,6\}$$

Define:

- $A = \{\omega_1 = 1\}$ (first roll is 1)
- $B = \{\omega_2 = 2\}$ (second roll is 2)

Now:

- $A, B \subseteq \Omega$

Then:

$$P(A) = \tfrac{1}{6}, \quad P(B) = \tfrac{1}{6}, \quad P(A \cap B) = \tfrac{1}{36}$$

✔ **They are independent.**

Because

$$A \cap B = \{(1,2)\}$$

Quiz

In a classification system:

- Let A: "An email is classified as spam"
- Let B: "An email is classified as not spam"

Claim: A and B are independent events

A: True

B: False

Quiz

- **Event A**: You have **$100 left** at the end of the day.
- **Event B**: You **spend $10** during the day.

Are Events A and B disjoint (i.e., mutually exclusive)?

A: True

B: False

Independence

Union of Independent Events

If $P(A) > 0$ and $P(B) > 0$ **and independent**, then

$$P(A \cup B) \neq P(A) + P(B)$$

Because A and B are **not** disjoint!

But for independent $A_1, \ldots, A_n$, use De Morgan's law:

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = 1 - \prod_{i=1}^{n}(1 - P(A_i))$$

This formula **requires** the A_i's to be independent.

Proof:

$$\begin{aligned}
P\left(\bigcup_{i=1}^{n} A_i\right) &= 1 - P\left(\left(\bigcup_{i=1}^{n} A_i\right)^c\right) && \text{(complement rule)} \\
&= 1 - P\left(\bigcap_{i=1}^{n} A_i^c\right) && \text{(De Morgan)} \\
&= 1 - \prod_{i=1}^{n} P(A_i^c) && \text{(independence)} \\
&= 1 - \prod_{i=1}^{n}(1 - P(A_i)) && \text{(complement rule).}
\end{aligned}$$

Quiz

Suppose A, B, and C are **independent events** with

$$P(A) = 0.6, \qquad P(B) = 0.3, \qquad P(C) = 0.1.$$

What is

$$P(A \cup B \cup C)?$$

A: 0.018

B: 0.902

C: 0

D: 0.748

Independence

Example: What is the probability of **at least one crash**?

- The probability of dying in **one flight** is:

$$p_c = \frac{1}{4{,}000{,}000}$$

- You take **400 flights** over time.

Solution

Assume each flight is **independent** of the others.

Let:

- A_i: crash on flight i
- A_i^c: no crash on flight i

Use the Formula:

$$P\left(\bigcup_{i=1}^{n} A_i\right) = 1 - \prod_{i=1}^{n} (1 - P(A_i))$$

Since all flights have the same crash probability $P(A_i) = p_c$, and are **independent**, this simplifies to:

$$P(\text{At least one crash}) = 1 - (1 - p_c)^{400}$$

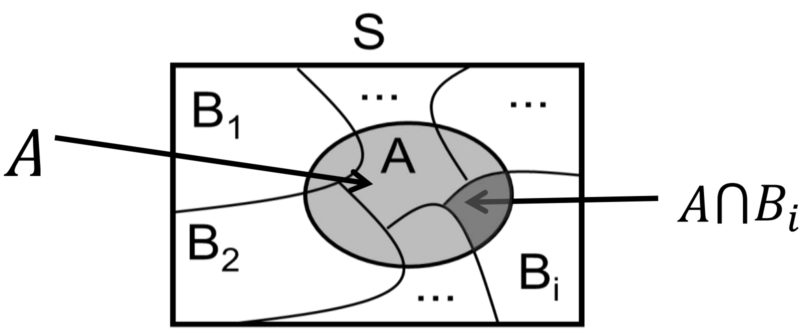
Result:

$$P = 1 - \left(1 - \frac{1}{4.000.000}\right)^{400} = \frac{1}{10000}$$

Total Probability

Suppose we want to compute the probability of some event A, but we can't find $P(A)$ directly.

However, we know how likely A is to happen **given** some other events $B_1, B_2, B_3, \ldots$, which split the entire sample space into disjoint parts. Then we can compute $P(A)$ by "adding up" its parts.



Law of Total Probability:

If $B_1, B_2, B_3, \ldots$ form a **partition** of the sample space S (i.e., they are disjoint and cover all of S), then for any event A:

$$P(A) = \sum_i P(A \mid B_i) \cdot P(B_i)$$

Proof

$$A = A \cap S = A \cap \left(\bigcup_i B_i \right) = \bigcup_i (A \cap B_i)$$

Since the B_i are disjoint, so are the $A \cap B_i$. Thus:

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A \mid B_i) \cdot P(B_i)$$

Example: Fault Detection in Production Lines

A factory has **three automated production lines** that each produce the same kind of component. Each line produces **100 components per day**:

- **Line 1**: 75 components are **flawless**, 25 have **minor defects**
- **Line 2**: 60 flawless, 40 with minor defects
- **Line 3**: 45 flawless, 55 with minor defects

Suppose that:

- A component is selected by first picking **one production line at random**,
- Then a **random component** is selected from that line.

What is the probability that the selected component is flawless?

Let:

- F = "component is flawless"
- B_i = "component comes from Line i"

We know:

$$P(F|B_1) = 0.75, \quad P(F|B_2) = 0.60, \quad P(F|B_3) = 0.45$$

Since each line is equally likely to be picked:

$$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$$

Now apply the **Law of Total Probability**:

$$P(F) = P(F|B_1)P(B_1) + P(F|B_2)P(B_2) + P(F|B_3)P(B_3)$$
$$= (0.75)(\frac{1}{3}) + (0.60)(\frac{1}{3}) + (0.45)(\frac{1}{3}) = 0.60$$

Bayes' Rule

Bayes' Rule helps us **reverse conditional probabilities**. That is, it tells us how to compute $P(B|A)$ when we know $P(A|B)$.

Let's start with the **definition of conditional probability**:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiply both sides by $P(B)$:

$$P(A \cap B) = P(A|B) \cdot P(B)$$

Now swap the roles of A and B:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Substitute the previous expression for $P(A \cap B)$:

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$$

Bayes rule:

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$$

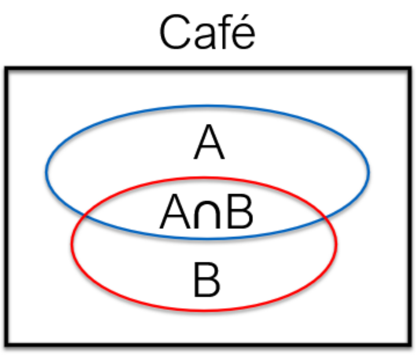
Bayes' Rule

Example: Café revisited

What is the probability that a customer orders coffee given they ordered cake?

Events:

- **A** = Customer orders **coffee**
- **B** = Customer orders **cake**



$\Pr(A) = 0{,}70$

$\Pr(B) = 0{,}40$

$\Pr(A \cap B) = 0{,}20$

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)} = \frac{0{,}20}{0{,}40} = \frac{1}{2}$$

Use **Bayes rule**:

$$Pr(B|A) = \frac{Pr(A|B) \cdot Pr(B)}{Pr(A)} = \frac{½ \cdot 0{,}40}{0{,}70} = \frac{2}{7} = 0{,}286$$

Bayes' Rule

Example: Particle Detector in a Physics Experiment

A particle detector is used to identify **Higgs bosons** in high-energy collisions. However, sometimes the detector **mistakes other particles** as Higgs bosons.

- Only **0.1%** of all collisions produce a Higgs boson.
- If a Higgs boson is produced, the detector correctly identifies it **95%** of the time.
- If no Higgs boson is produced, the detector still gives a **false positive 2%** of the time.

Now suppose the detector signals a **positive detection**.

What is the probability that a Higgs boson was actually produced in that collision?

Let:

- H: A Higgs boson was produced.
- T: The detector gives a positive signal.

Using **Bayes' Rule** and the **Law of Total Probability**:

$$P(H \mid T) = \frac{P(T \mid H) \cdot P(H)}{P(T \mid H) \cdot P(H) + P(T \mid \text{no } H) \cdot P(\text{no } H)}$$

Substitute known values:

- $P(H) = 0.001$
- $P(\text{no } H) = 0.999$
- $P(T \mid H) = 0.95$
- $P(T \mid \text{no } H) = 0.02$

$$\Rightarrow P(H|T) \approx 0.0454$$

Engelsk – dansk oversættelse

Conditional probability	Betinget sandsynlighed
Chain rule	Kædereglen
Bayes' rule	Bayes' regel
Total probability	Total sandsynlighed
Partition of sample space	Opdeling af udfaldsrum
Disjoint events	Disjunkte hændelser
Independent events	Uafhængige hændelser
Complement rule	Komplementreglen
Inclusion-exclusion principle	Inklusions-eksklusionsprincippet